

● AAROGYA GRID

The medicine is already there. Nobody can see it.

National medicine supply intelligence for India's primary health network — forecasting stock-outs across 2,816 facilities, and finding the stock sitting an hour away that would prevent them.

Build with AI: **Code for Communities**

Problem statement 03 — Smart Health & Supply Chain Resilience

Built with **Google Gemini**

THE PROBLEM

A sub-centre runs out of anti-snake venom in monsoon. The vials exist.

They are ninety minutes away, in a facility that will write them off unused in six weeks. Nobody knows either fact.

Both facilities report into a paper register that reaches the district office weeks later, if at all. India's public health supply chain does not primarily fail on procurement volume — it fails on **visibility**, and on **lateral movement**: the ability to see a shortage forming and move stock sideways before it becomes a stock-out.

4,594

POSITIONS AT ZERO STOCK

5,471

CRITICAL POSITIONS

1.92 M

UNITS EXPECTED SHORTFALL

₹11.4 L

STOCK HEADING TO EXPIRY

WHAT IT DOES

Four steps, from a spoken sentence to a signed dispatch order.

01

Capture

An ANM speaks, in Hindi, or photographs a paper register. Gemini turns it into structured stock data.

02

Forecast

Croston's method for intermittent demand, corrected for the censoring that stock-outs create in the ledger.

03

Rank risk

Monte Carlo over the resupply window returns stock-out probability and expected shortfall — not a point estimate.

04

Redistribute

Pairs facilities heading for expiry with facilities heading for zero. Names the batch. Prices the trip.

2,816

FACILITIES

128

DISTRICTS · 16 STATES

80,896

FACILITY × DRUG POSITIONS

232 M

PEOPLE IN CATCHMENT

THE FORECASTING PROBLEM NOBODY ACCOUNTS FOR

A stock ledger records what was dispensed – not what was needed.

Once a facility hits zero, demand keeps arriving and stops being recorded. Fit naively on that ledger and you **systematically under-forecast exactly the facilities that are already failing** – then allocate them less, so they stock out again.

We fit with stocked-out periods excluded. Measured against the same estimator run on uncensored demand, so seasonality and recency cancel and only the censoring effect remains.

DISTRICT TIER	PAIRS	NAIVE	CORRECTED
Functioning	2,551	-2.7%	-0.8%
Strained	1,018	-4.7%	-0.8%
Disrupted	301	-10.0%	-2.6%

7.4 percentage points of forecast bias recovered in the worst-served districts — the ones where being wrong costs the most.

Not a dashboard alert. A dispatch order a storekeeper can act on.

GENERATED DISPATCH ORDER • BHAGALPUR → PURNIA, BIHAR

CHC Purnia-01 holds **0 sachets** of Oral Rehydration Salts — a **Vital** drug — against a **100%** chance of running short inside its 7-day resupply window. **SC Bhagalpur-10**, in the next district, can spare **171**. Moving them **20 km** costs **₹810** and averts an expected **118.7 sachets** of unmet demand, cutting stock-out risk from **100% to 8%**.

pick list: B007-SC010 × 87 (exp 2027-01-02) • B006-SC010 × 84 (exp 2028-12-26) • risk reduction 92 pp

Every recommendation names the **specific batches and their expiry dates**, oldest first, and the quantity is bounded by what those batches physically hold. A recommendation a storekeeper cannot fill is not a recommendation. This one also crosses a district boundary — the clause the brief adds this edition — and Purnia is one of the eight worst-scoring districts in the country on our own ranking.

Visibility was not the binding constraint. A truck per tablet was.

Of the needs the optimiser could not serve, we expected most to fail because the stock simply wasn't there. That is not what happened. **98.8% failed the benefit/cost gate** — the medicine existed, a median **27.1 km** away. **Not one** failed for want of donor stock anywhere.

The cause was in our own pricing. The planner solves one drug at a time, so two orders on the same route were computed in different calls and never met: **2,798 orders were billed 2,798 vehicles for 2,083 distinct routes**. And every order stayed inside its own district — not because the optimiser refused, but because nobody had ever handed it two.

So we priced the route once and opened the boundary. They are one change: a cross-district trip is longer, fails the same gate harder, and only clears once several orders share the vehicle.

NATIONAL PLAN	BEFORE	AFTER
Dispatch orders	2,798	7,097
Vehicle trips	2,798	2,449
Crossing a district	0	743 trips
Needs declined	14,537	8,879
No stock anywhere	217	0

3,873 of those orders could not justify a vehicle alone. They are filled for the price of handling, because one was already going — stock-outs that were being refused for a truck nobody needed to hire.

Both columns summed from economics.reasonHistogram across the 128 shipped payloads · the median distance is over the declined needs those payloads list · before = commit 73cf60a

Redistribution does not pay for itself in cash. It pays in lives.

NATIONAL PLAN	VALUE
Waste averted	+ ₹3.46 L
One vehicle per order	— ₹87.4 L
Transport cost, consolidated	- ₹36.4 L
Net cash position	- ₹32.9 L
Shortfall averted	885,946 units

6,851 dispatches on 2,520 vehicle trips across 128 districts. Against the previous build:
79% more shortfall averted for 9% more cash.

We could have shown a ₹2.7 crore headline. It would have been a shadow price on averted shortfall doing 112% of the work.

Instead the multiplier is **an explicit dial**. The plan breaks even when one unit of averted unmet demand is valued at **₹3.72** — down from ₹6.11, because consolidation moved the same medicine for less. Whether a dose of a *Vital* medicine reaching a patient is worth that is a **ministry's judgement, not an engineer's** — so we surface the parameter rather than bury it in a headline.

The officer asks in Hindi. The model calls the tools. Every number is quoted, never computed.

LIVE • DISTRICT HEALTH OFFICER ASKS

"Bastar mein kaun se centre par dawa khatam hone wali hai?"

```
1 list_positions {limit:10} → 10 rows
2 district_status {} → 1 row
3 drug_reference {Ceftriaxone} → 1 row
4 drug_reference {Artesunate} → 1 row
answered from 4 calls • 7 facilities and 3 drugs cited • 22.2s
```

Ten tools, all pure functions over already-computed state. The **tool-call trace is shown in the interface** — a grounded answer nobody can check is indistinguishable from a confident guess.

NO IDS The model is never given a district code, facility id or drug id to emit. Asked about a district it had not been handed, it **invented the code "Lucknow"** — so arguments are natural-language names, resolved deterministically inside the tool.

NO MATHS It quotes numbers, never derives them. Testing caught it turning a 0.998 probability into "99.8%" — correct arithmetic, but **every ratio now ships a precomputed percent** to quote instead.

HAND LOOP The SDK's automatic function calling is deliberately unused: we reject unknown tool names, **Zod-validate every argument** before it touches data, and record the trace ourselves rather than asking the model what it did.

AUDITED Every numeric token in the model's prose is matched against the payloads it actually saw. The audit is **proven able to fail** by tampering a known-good answer. Result: **zero hallucinated numbers** across 8 scenarios.

MEDICINES ARE ONE OF THREE THINGS A FACILITY RUNS OUT OF

Beds and people, on the same screen as the stock – because they explain it.

WORKFORCE	POSTS
Sanctioned	33,920
In position	26,981
Present today	22,232
Specialist vacancy	45.5%

Three levels, not one number. In rural India the **vacancy** gap and the **absence** gap are different problems with different fixes, and the distance between them is the finding.

That last number is the hinge. It does not rewrite any measured quantity — it **widens the error bar** around it. Three resources, one system.

BEDS	COUNT
Sanctioned	37,888
Functional	33,017
Staffable today	31,546
Occupancy · at capacity	83.3% · 100

219 stock-holding facilities have no pharmacist in post to keep the register. 393 facilities, covering 8.5 million people, carry stock figures nobody is in position to verify.

Four users, one dataset, four different questions.

ANM SUB-CENTRE	Speaks one sentence into a phone instead of filling a register by hand. Told immediately if what she reported looks wrong.
Storekeeper PHC / CHC	Receives a dispatch order naming a batch, a quantity, an expiry date and a distance. Nothing to interpret.
District Health Officer DISTRICT	Asks the grid a question in their own language and gets a grounded answer with its working shown. Sees what will fail in two weeks, and what can be moved to prevent it.
State & ministry NATIONAL	Sees which districts are structurally failing rather than unlucky, and where a logistics contract or a recruitment drive would buy more than more procurement.

Three adapters stand between this and real data. The country scales linearly.

Everything downstream consumes one interface, FacilityDrugState. Three adapter functions produce it, and they are the only code a real deployment replaces — a **DVDMS / e-Aushadhi** stock extract, an **ABDM Health Facility Registry** pull, and an **HMIS** bed and attendance feed. The forecaster, the optimiser, the batch and every screen are **untouched**.

THE INTEGRATION SEAM

```
sim/facilities.ts → ABDM HFR
sim/inventory.ts → DVDMS extract
sim/resources.ts → HRMIS / HMIS
```

```
forecast/ · optimize/ · ai/ · app/ unchanged
```

SCOPE	DISTRICTS	NIGHTLY BATCH
This demo, measured	128	~3 min
One large state	~40	~60 s
All India, demo density	780	~20 min

Simulation and forecasting stay **embarrassingly parallel** — seeded per (seed, facility, drug). Planning is not: donor stock can be promised **exactly once**, so the run shares one allocation ledger. Two districts plan concurrently only when their clusters are disjoint — **9 rounds**, not 128 tasks. And sharding by state is **not** clean: **78 of 128** clusters cross a state line, the same fact that produces 74 cross-state corridors.

Running on **Cloud Run, asia-south1**. Compute and inference are both in Mumbai and the container holds **no credential at all** — the service account attaches to the service. Nothing to rotate, leak, or revoke.

Stated plainly, because the honest answer is the stronger one.

REAL

Districts, their coordinates and state assignments. Facility tier structure, **bed norms and the sanctioned staffing establishment** per IPHS 2022. A 47-item catalogue — **46 drugs from India's National List of Essential Medicines** plus one tracked consumable — with VED classification, pack units, cold-chain flags and unit costs.

Real PHC inventory data is not public. Rather than claim otherwise, we built the seam that makes the swap a one-file change — and **every number in this deck is reproducible from a seed**, by anyone, from the public repository.

SIMULATED

The stock ledger, bed occupancy, staff attendance — **and district catchment populations**. All from a seeded, deterministic simulator parameterised from IPHS norms and published seasonality, **not fitted to observed data**. District and state rankings reflect a synthetic reliability parameter, not real performance.